A

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 341: Microprocessors Semester: Summer 2026

Examination: Quiz 4 Duration: 30 Minutes Full Marks: 20

Name: _____ ID: _____ Section: _____

Question 1 [10 marks]

	Instruction	**Answer**
i)	`IN AX, 2A40H`	
ii)	`MOV DX, 2A40H` `IN AX, DX`	
iii)	`MOV AX, [2A40H]`	

a) Identify **addressing technique** of each instructions mentioned above and fill the **Answer** column. Justify your answer based on the **8086 instruction format** and **addressing constraints. [2]**

b) *"The I/O instruction technique showcased in the 3rd instruction can access a greater number of ports compared to the one in the 2nd instruction."* - Critically evaluate the statement with proper technical reasoning. **[2]**

c) A Computer System uses a modified DMA controller named **DMA6,** which supports **6 DMA channels** instead of the conventional 4-channel architecture of the 8237 DMA controller.

   i) Construct a **labelled diagram** showing how **two secondary DMA6** controllers can be interfaced with one primary DMA6 controller in **cascading mode** using **appropriate pins. [3]**

   ii) Explain the **sequence of operations** involved during **DMA-based data transfer** between the Intel 8086 microprocessor and the **cascaded DMA6 controllers. [3]**

Question 2 [10 marks]

Address	Data
0006CH	00H
0006DH	03H
0006EH	00H
0006FH	05H
0009CH	00H
0009DH	07H
0009EH	00H
0009FH	00H
00108H	00H
05300H	89H
05301H	8BH
00700H	2BH
00701H	90H

a) Use the provided table to calculate the **starting address** of the ISR for interrupt **type 27.** Also, deduce the **first 2-byte hex instruction** that will be fetched to the instruction queue from the ISR. **[3]**

Starting Address	
2-byte hex instruction	

b) Explain if the interrupt caused by the `INT 27` instruction is **maskable or not.** Provide your reasoning. **[3]**